EXPERIMENT 13

Host Discovery Tools & Techniques (Ping Sweep, ARP Ping, fping)

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AIM:

To execute and demonstrate the functionality, syntax, and outputs of host discovery tools and network sweep techniques in Kali Linux Operating System, including Nmap ICMP ping sweep (-sn), local subnet ARP discovery (-PR), and fping.

COMMANDS AND DESCRIPTION:

1. Nmap Ping Sweep (nmap -sn):

• *Description:* Performs host discovery across an entire IP subnet without conducting any port scans. Sends ICMP Echo Requests, TCP SYN to port 443, TCP ACK to port 80, and ICMP Timestamp requests to detect live hosts.
• *Syntax:* sudo nmap -sn <subnet_CIDR> (e.g., 10.0.2.0/24)
• *Characteristics:* Quickly maps out active devices across large address spaces without triggering full port scan alerts.

2. Nmap ARP Discovery (nmap -sn -PR):

• *Description:* Uses raw Address Resolution Protocol (ARP) frame requests to identify active hosts on the local Ethernet/WiFi subnet. It is faster and far more reliable than IP ping because local OS firewalls cannot block ARP queries without losing network connectivity.
• *Syntax:* sudo nmap -sn -PR <target_IP/subnet>
• *Scope:* Applicable exclusively to local broadcast domains (Layer 2).

3. fping Utility (fping -a -g):

• *Description:* A high-performance command-line ping tool that sends round-robin ICMP requests to multiple hosts simultaneously, generating a concise list of reachable live hosts.
• *Syntax:* fping -a -g <start_IP> <end_IP>
• *Options:* -a (show alive hosts only), -g (generate IP target range).

OUTPUT:



RESULT:

Thus, host discovery tools and techniques including Nmap ping sweep (-sn), ARP-based local discovery (-PR), and fping subnet scanning were successfully executed in Kali Linux Operating System, verifying active IP mapping, live host detection, and network asset identification.